

Energy, Work, and Power of the Body

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Objective

Students will understand the concepts of energy expenditure in the human body, including basal metabolic rate (BMR), energy conservation, and the role of different organs in energy consumption.

They will also explore real-life applications of these concepts in relation to physical activity and metabolic processes.

All body activities including thinking, doing work, or keeping the body temperature constant involve energy.

For example under resting (basal) conditions:

- The skeletal muscle and the heart using (25%) of the body's energy
- 19% percent in the brain
- 10% in the kidneys
- and 27% being used by the liver and the spleen.
- A small percent of about 5% of food energy being excreted in feces and urine.

Conservation of energy

Change in the stored energy = Heat lost + Work done.

$$\Delta u = \Delta Q + \Delta w$$

Where: -

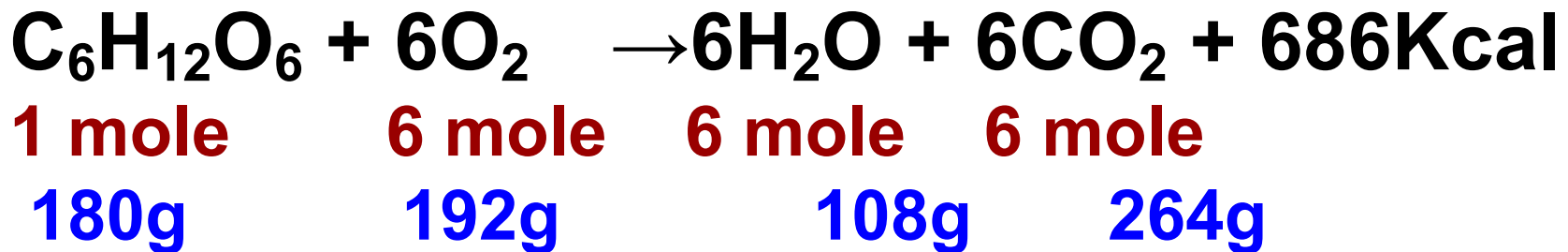
- * Δu is the change in the internal energy.
- * ΔQ is the change of quantity of heat of the system.
- * Δw is the work done.

Energy changes in the body

The units are joule or calorie **1cal= 4.184J** or **1Kcal=4184J**

The power is defined as energy :
or work per unit time = **J/s = watt.**

In the oxidation process within the body heat is produced as energy of metabolism. The rate of oxidation is called metabolic rate. For example the oxidation of one mole of glucose can be shown as: -



When the body is completely at rest. It will have the lowest rate of energy consumption this is called the basal metabolic rate (BMR) .

$$\text{BMR} = 50 \text{ kcal} / (1 \text{ m}^2 \text{ of the body surface area. hr.}) = 1 \text{ met}$$
$$1 \text{ Kcal} / \text{hr} = 1.162 \text{ W}$$

What is the quantity of BMR for the body that has surface area = 1.4 m^2

- **BMR** depends on **sex**, age, height, and weight.
- **BMR** depends primarily upon **thyroid function**.
Overactive thyroid gives higher (**BMR**).

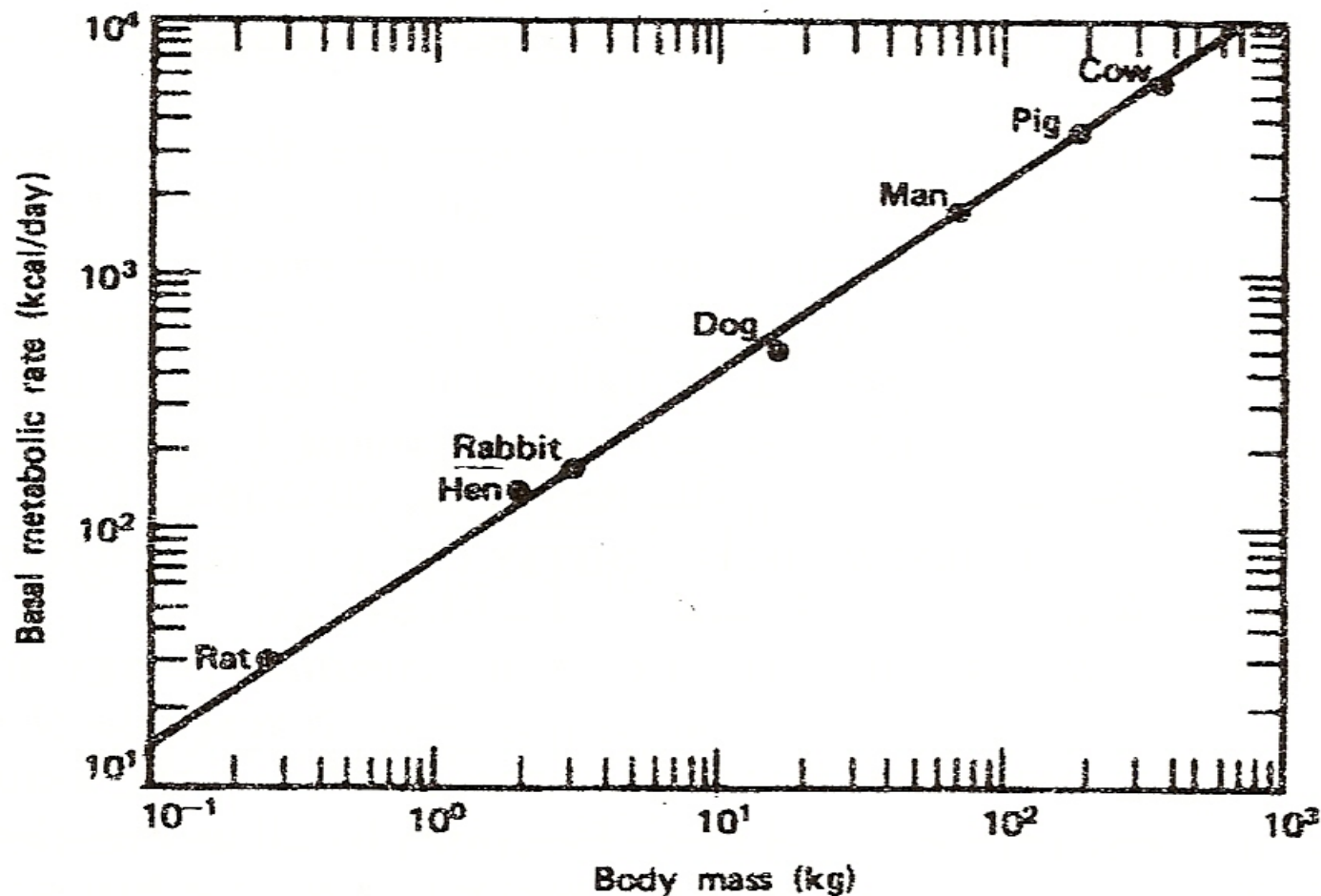


Figure 5.1. Relationship between the basal metabolic rate and the body mass for several different animals.

The **BMR** depends to a large extent on the body temperature for an increase of 1°C it will change by **10%** so for 3°C the change will be **30%** this is applicable for the decrease in the body temperature, for this reason hibernating animals at low body temperature will reduce the metabolic rate very much.

Why hibernating animals (like frogs) at low temp. during winter?

(BMR) is sometimes determined from O_2 consumption.

Activity	O ₂ Consumption (liters/min)	Equivalent Heat Production		Energy Consumption (mets—50 kcal/m ² hr)
		kcal/min	W	
Sleeping	0.24	1.2	83	0.82
Sitting at rest	0.34	1.7	120	1.15
Standing relaxed	0.36	1.8	125	1.25
Riding in automobile	0.40	2.0	140	1.35
Sitting at lecture (awake)	0.60	3.0	210	2.05
Walking slow (4.8 km/hr)	0.76	3.8	265	2.60
Cycling at 13–17.7 km/hr	1.14	5.7	400	3.90
Playing tennis	1.26	6.3	440	4.30
Swimming breaststroke (1.6 km/hr)	1.36	6.8	475	4.65
Skating at 14.5 km/hr	1.56	7.8	545	5.35
Climbing stairs at 116 steps/min	1.96	9.8	685	6.70
Cycling at 21.3 km/hr	2.00	10.0	700	6.85
Playing basketball	2.28	11.4	800	7.80
Harvard Step Test ^a	3.22	16.1	1120	11.05

^a Adapted from P. Webb, in J. F. Parker and V. R. West (Eds.), *Bioastronautics Data Book*, National Aeronautics and Space Administration, Washington, D.C., 1973, pp. 859–861.

^a A test in which the subject steps up and down a 40 cm step 30 times/min for 5 min.

Work and Power

Mechanical work is usually defined by $\Delta w = F \Delta x$ when F is the force on the same line of displacement x or it can be also written as:-

$$\Delta w = F \Delta x \cos \theta$$

where θ is the angle between (F) and the direction of movement the power is work per unit time.

$$P = \Delta w / \Delta t = F \Delta x / \Delta t = Fv$$

Where v is the velocity

When the force is perpendicular to the displacement work will be zero such as a walking body, his weight is perpendicular to distance of movement but practically it will not be zero because he uses energy against friction and other movements of his body, but in the case of climbing person for distance (h) the weight is on the same line of displacement then the **work = mgh**,

The efficiency of human body is $E = \text{work done} / \text{energy consumed}$.

For physical activity the efficiency can be increased by 20% by training such as trained person for cycling.

The maximum work capacity of the body is variable for short period of time it can perform very high power level, (like running very fast) but it is more limited for longer periods. It is found that long term power is proportional to the maximum rate of oxygen consumption in the working muscle.

5.4. Heat Losses from the Body

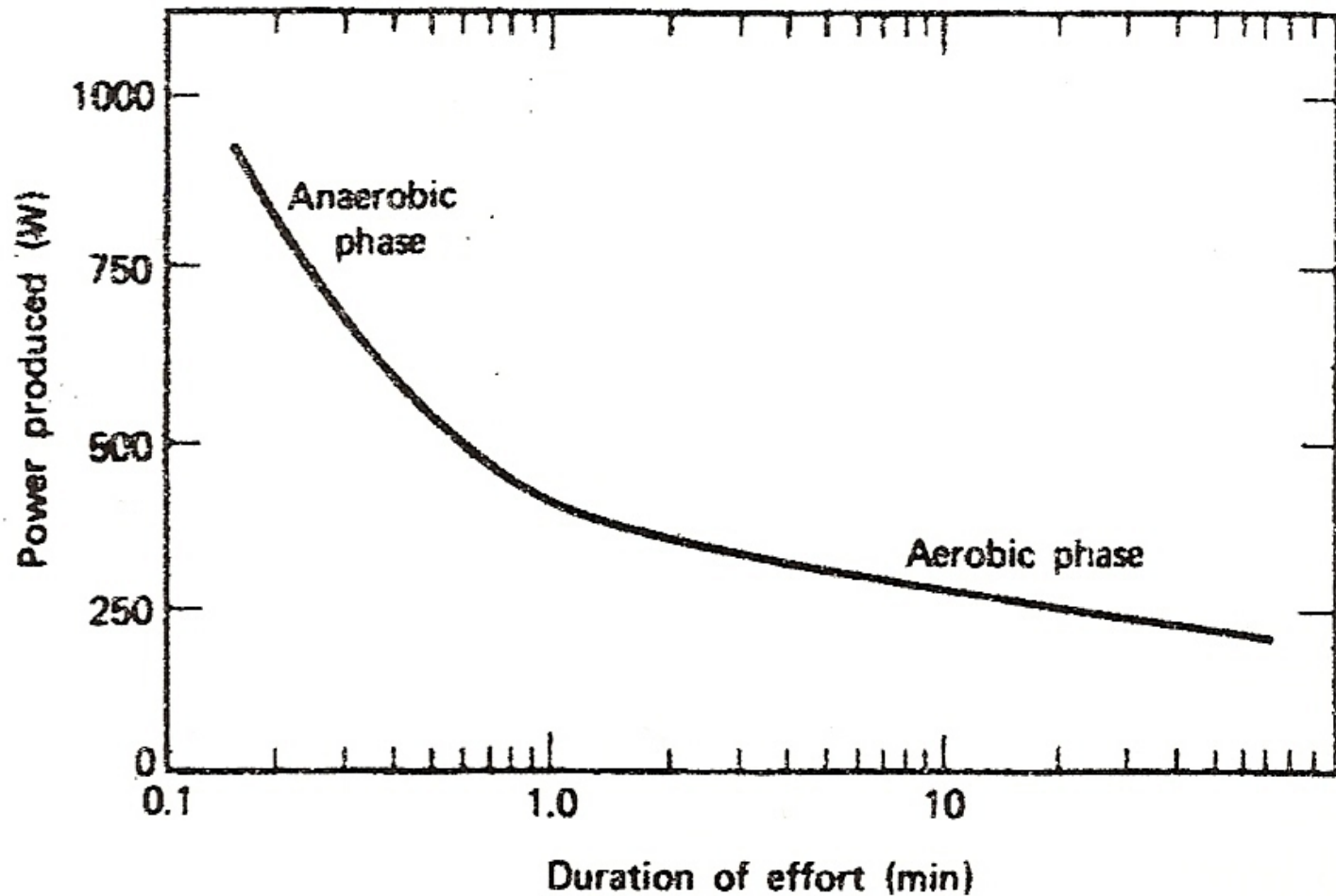


Figure 5.3. Typical power output on a bicycle versus duration of effort for an average healthy adult. Anaerobic work can only be maintained about 1 min.

1. What percentage of the body's energy is used by the brain under resting conditions?

- a. 19%
- b. 27%
- c. 10%
- d. 5%

2. What is the formula for calculating the change in stored energy?

- $\Delta u = \Delta Q + \Delta w$
- $\Delta u = \Delta w - \Delta Q$
- $\Delta u = \Delta Q - \Delta w$
- $\Delta u = \Delta w + \Delta Q$

3. Which of the following units is NOT used for energy?

- Joule
- Calorie
- Watt
- Kilogram

4. What is the basal metabolic rate (BMR) for a body with a surface area of 1.4 m²?

- 70 kcal/hr
- 100 kcal/hr
- 50 kcal/hr
- 60 kcal/hr

5. Which factor does NOT affect the basal metabolic rate (BMR)?

- Age
- Height
- Weight
- Dietary habits

6. What is the primary function that influences BMR?

- Physical activity
- Thyroid function
- Body temperature
- Metabolic rate

7. Why do hibernating animals experience a reduced metabolic rate during winter?

8. How is mechanical work defined in the context of physical activities?

9. What is the relationship between the maximum work capacity and oxygen consumption?

10. Explain the fig.

